Noah Bean

noahbean3396@gmail.com — (971) 419-0100 — linkedin.com/in/noah-bean — github.com/noahbean33

Education

Oregon State University Corvallis, OR
B.S. Electrical and Computer Engineering GPA: 3.81/4.0

Technical Skills

- **Programming Languages:** C, C++, Python, Bash, Assembly (ARM, RISC-V, x86)
- **Microcontrollers & Architectures:** STM32, ESP32
- **RTOS & Operating Systems:** FreeRTOS, Bare-Metal, Embedded Linux (Buildroot)
- **Communication Protocols & Interfaces:** UART/USART, SPI, I^2C, Modbus, MQTT, Bluetooth/BLE, CAN, TCP/IP, UDP
- **Peripherals, Drivers & Low-Level:** GPIO, Timers/PWM, ADC/DAC, DMA, Interrupts, CMSIS register-level drivers, Sensor interfacing, Actuator control, Power management & low-power modes, Clock/PLL configuration, OTA updates, Bootloaders
- **Development Tools & Build Systems:** STM32CubeIDE, ESP-IDF, GCC/ARM GCC toolchain, CMake, Makefiles, Git, JIRA
- **Debugging, Testing & Instrumentation:** GDB, OpenOCD, JTAG/SWD/ST-Link, Oscilloscopes, Logic Analyzers, Protocol Analyzers, Unit testing, Static analysis (Cppcheck)

Professional Experience

Collins Aerospace Wilsonville, OR
Systems and Electrical Engineering Intern June 2025 – December 2025

- Developed Python-based automated test scripts and frameworks to standardize functional and electrical verification of avionics hardware, significantly improving data collection consistency using oscilloscopes, power supplies, and lab instrumentation.
- Implemented and verified real-time digital signal processing algorithms for sensor data acquisition on avionics units, ensuring reliable performance in embedded environments.

Intel Corporation Hillsboro, OR
Electrical Engineering Intern April 2024 – September 2024

- Performed root-cause analysis and electrical debug on server-grade CPUs, GPUs, and DDR5 memory subsystems using oscilloscopes and logic analyzers; isolated signal integrity issues to support high-volume manufacturing reliability.
- Engineered Python frameworks to automate extraction and analysis of on-chip telemetry data and hardware logs, accelerating failure analysis workflows for silicon validation.

Projects

Air Quality Monitor STM32F446RE • FreeRTOS • Modbus RTU *C, CMSIS*

- Architected a production-grade indoor air quality monitoring system on STM32F446RE using FreeRTOS with five concurrent tasks for sensor acquisition, system health, error handling, Modbus communication, and data logging.
- Implemented custom Modbus RTU slave protocol, interrupt-driven communication, synchronization primitives (queues, semaphores, mutexes), and watchdog timer for robust, fault-tolerant embedded operation.

ESP32 IoT Weather Station ESP32 • ESP-IDF • AWS IoT Core • FreeRTOS *C*

- Engineered a full-featured IoT weather station with DHT22 sensing, dual-mode Wi-Fi (AP + Station), and MQTT publishing of environmental data to AWS IoT Core.
- Designed and implemented a responsive HTTP web server supporting OTA firmware updates, real-time dashboard, NVS credential storage, SNTP time synchronization, and RGB LED status indicators.

Custom Real-Time Operating System STM32 • Bare-Metal *C, ARM Assembly*

- Built a complete bare-metal RTOS kernel from scratch for STM32 microcontrollers, including thread control blocks, PendSV-based context switching, and hardware abstraction layers for GPIO, UART, timers, and ADC.
- Implemented multiple scheduling policies (round-robin, cooperative, periodic, priority-based preemptive) with full thread lifecycle management and state tracking.